

Kayden Pellegrini

Data Engineer | AI Systems Developer

Edenvale / Johannesburg, South Africa | 064 511 5136 | developer.kayden@gmail.com | linkedin.com/in/kaydenpellegrini
github.com/KaydenPellegrini | kayden.co.za

Italian and South African citizen. Full EU work authorisation, no sponsorship required.

PROFESSIONAL SUMMARY

On the data side I work on modelling, cost and margin at the grain of the individual unit, reconciliation between systems that disagree with each other, and tests that fail the build when a number stops being trustworthy. On the AI side I work on LLM tooling connected to live business systems through Model Context Protocol, audit skills that verify their own fixes, and a written set of standards for what AI-assisted analysis is allowed to claim.

I do this for a medical device distribution company in Johannesburg, where I also build the internal applications and reporting that operations runs on. That covers procurement, inventory, stock movement, stocktaking and receiving in Power Apps and Dataverse, reporting and forecasting in Power BI, and the Power Automate workflows that connect those to finance, printing and email. Around that I work with RFID and barcode identification, REST APIs, secure internal web pages and broadcast AV infrastructure.

KEY SKILLS

Data Engineering: dbt, DuckDB, SQL, Python, ETL, Data modelling, Data quality testing, Reconciliation, Forecasting, Git, GitHub Actions, CI/CD

AI: Model Context Protocol, LLM tooling, Prompt engineering, Agent evaluation, Claude Code skills

Power Platform and BI: Power Apps, Power Automate, Dataverse, Power BI, DAX, Excel

Languages and Frameworks: Python, TypeScript, Swift, Java, Next.js, React, Three.js and WebGL, Android development, HTML, CSS

Systems and Integration: REST APIs, JSON, Sage Accounting, PrintNode, SharePoint, Microsoft Entra ID, Cloudflare Access, AWS EC2

Identification and Hardware: RFID, SGTIN-96 and EPC decoding, GS1 barcode and DataMatrix, Bluetooth Low Energy, Barcode systems, Mobile scanner integration

Delivery and Broadcast: System testing, Deployment, Troubleshooting, Networking, vMix, NDI

EXPERIENCE

Data and Systems Developer

Virtumed (Pty) Ltd, Johannesburg, Gauteng | April 2024 to Present

- Design and maintain the data flows that move transactional data between Sage Accounting, Dataverse, internal web applications and the reporting layer, using REST APIs, Power Automate and Python.
- Model the operational data in Dataverse, including the table relationships and the serial level traceability that links inventory records to purchase order lines and their actual cost.
- Reconcile and rebuild business datasets before they are used for analysis, including a 6,984 line purchases export across 145 suppliers cross checked against a separate item level report, and a 16,597 row inventory register merged from two misaligned source tables.
- Build cost and margin models from source transaction data, reconstructing landed cost from supplier price, actual exchange rate, freight, customs and bank charges rather than list price or valuation figures.
- Apply data quality controls as standard practice, covering join validation before results are trusted, duplicate and truncation checks, grain control on serialised records, and explicit separation of verified results from assumptions in every output.
- Write Python for data processing and validation with reproducibility in mind, keeping source files untouched, logging every transformation applied, and proving logic on a slice before running it at full scale.
- Designed and built a multi agent verification framework for AI generated work, using independent review and confirmation passes followed by an adjudication pass that checks each change against the actual result and classifies it as verified, overclaimed, hallucinated, cosmetic or a regression.
- Integrated LLM tooling into the Microsoft stack through Model Context Protocol across Dataverse, Power Apps and Microsoft 365.

- Set the internal standards for AI-assisted analysis, covering one source of truth per data domain, confidence labelling, read-only scoping on live systems and human confirmation before destructive operations.
- Built reusable validation and analysis modules and shared them with colleagues.
- Build Power BI dashboards and reports covering sales, inventory, stock risk, sales activity, locations and product groups, with forecasting and comparative analysis from historical company data.
- Design, develop and maintain the internal systems for procurement, inventory, stock movement, stocktaking, receiving and reporting in Power Apps, Power Automate and Dataverse, including the Sage Accounting and PrintNode integration that processes and prints delivery notes.
- Built an RFID stocktake system using handheld RFID scanners, Power Apps, Dataverse and Power Automate, with automated HTML email and printable stocktake reports.
- Developed a native iOS capture application in Swift that decodes SGTIN-96 EPC tags and GS1 barcodes over Bluetooth Low Energy and feeds serials directly into the stocktake app, cutting a 30 item scan to under 15 seconds.
- Act as technical lead for a cross site medical broadcast system used during live clinical cases, including provisioning and running an AWS EC2 relay in the Cape Town region as the low latency endpoint linking sites, alongside vMix production, NDI and SRT transport and network design.

Escape Room Manager

Hashtag Escape, Johannesburg, Gauteng | February 2022 to March 2024

- Managed daily escape room operations and bookings.
- Managed and troubleshoot room equipment, software and technical systems.
- Diagnosed faults between sessions.
- Supported customers and daily operations.

Waiter

The Fat Ginger, Johannesburg, Gauteng | June 2021 to January 2022

- Customer service and daily floor operations in a busy restaurant.

EDUCATION

Diploma in Systems Development

Boston City Campus | 2021 to 2023 | Certificate received 2024 | NQF Level 6

Relevant study: Java, Database Programming, SQL, Android Application Development, Systems Analysis and Design, Workflow Management, Software Testing, Technical Systems and Support, Systems Development Project Management

National Senior Certificate

Jeppe High School for Boys | 2016 to 2020

SELECTED WORK

Serial Level Margin and Purchase Reconciliation: A dbt project that rebuilds gross margin at the grain of the individual serialised unit, from purchase order to invoice, on synthetic data. Repository: github.com/KaydenPellegrini/serial-margin-dbt

Readiness Audits: Two Claude Code audit skills that verify their own fixes, one for AI agents and one for normal codebases. Repository: github.com/KaydenPellegrini/readiness-audits

RFID Stocktake and Inventory System: Handheld RFID scanning into Power Apps and Dataverse, with confirmed, missing and unexpected stock reconciled in one pass.

Business Intelligence and Forecasting: Power BI reporting across sales, inventory, stock risk, locations and product groups, with forecasting from historical data.

Sage and Print Automation: Power Automate moving operational delivery information into Sage Accounting and out to a physical printer through PrintNode.

Secure Internal Operations Tools: Internal web pages for stock movement, receiving and business workflows, published behind Cloudflare Access and Microsoft Entra.

Cross Site Medical Broadcast: Low latency audiovisual communication between doctors at separate clinical sites during medical cases.

LANGUAGES

English (fluent) | Afrikaans (moderate)